

Haley A. Oleynik

PhD Candidate

Institute for the Oceans and Fisheries, University of British Columbia

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EDUCATION

Colby College, Waterville, Maine, B.A. Biology & Sociology 2015

University of Delaware, Lewes, Delaware, M.S. Marine Science 2020

University of British Columbia, Vancouver, BC, PhD Fisheries Science in progress

RESEARCH EXPERIENCE

PhD Candidate Sept 2021 - Present

Institute for the Oceans and Fisheries University of British Columbia

Population modeling to identify drivers of productivity declines in Fraser River salmon and steelhead populations, with the goal of furthering understanding of regional, multi-species population patterns, identifying ecological and anthropogenic drivers of salmon population decline, and incorporating ecosystem information into fisheries and management assessment

Skills: Statistical modeling, Bayesian modeling, R programming, Git/Github, stakeholder engagement

Graduate Research Assistant Jan 2024 - Present

United States Army Corps of Engineers

Part time research analyst conducting statistical modeling of salmon passage and survival through dams in the Willamette River system in Oregon

Skills: Statistical modeling, Bayesian modeling, JAGS programming, data analysis, stakeholder engagement, R programming

Research Analyst Feb 2026 - May 2026

Pacific Salmon Commission Vancouver, B.C.

Analyzed salmon monitoring data to statistically evaluate the use of a new monitoring method for salmon passage through Hells Gate, a narrow gorge which poses passage challenges for salmon migrating up the Fraser River to their spawning grounds

Skills: Statistical modeling, R programming, Git/Github

Aquatic Biologist II Jun 2025 - March 2026

Department of Fisheries and Oceans Canada Vancouver, B.C.

Use empirical Harbour Seal diet data and modeled diet data from Ecopath with Ecospace ecosystem models to statistically validate model outputs and inform field sampling programs

Skills: Spatio-temporal modeling, ecosystem modeling, TMB programming, R programming

M.S. Student

Sept 2018 - July 2020

School of Marine Science and Policy University of Delaware

Analyzed data from the Delaware state trawl survey using multivariate statistical methods in R to describe the fish and invertebrate community in Delaware Bay, identify physical drivers of community composition, and evaluate long-term changes in the community

Skills: Fishery independent data analysis, multivariate statistics, timeseries analysis, R programming

Undergraduate Senior Honors Thesis

Sept 2014 - May 2015

Department of Sociology Colby College

Collected and analyzed quantitative and qualitative data to understand the societal impacts of the shifting lobster population and subsequent management on the lobster fishing industry in Maine

PROFESSIONAL EXPERIENCE**Teaching Assistant**

Sept 2023 - Present

Master of Data Science Program University of British Columbia

Teaching assistant for the following MDS courses: - DSCI 552: Statistical Inference and Computation I - DSCI 561: Regression I - DSCI 562: Regression II - DSCI 554: Experimentation and Causal Inference

Teaching Assistant

Spring 2026

Ecosystem Modeling with EwE University of British Columbia

Teaching assistant for a graduate level Ecosystem modeling course, in the Institute of the Oceans and Fisheries, which taught the use of Ecopath with Ecosim for marine ecosystem modeling.

Graduate Writing Consultant

Sept 2022 - April 2024

Center for Writing and Scholarly Communication University of British Columbia

Provided written and in-person consultations to graduate and undergraduate students on written work including class assignments, manuscripts, and dissertations

NOAA Sea Grant John A. Knauss Marine Policy Fellow

Feb 2021 - Feb 2022

National Stock Assessment Program NOAA Fisheries

Supported the National Stock Assessment Program in the Office of Science and Technology by organizing a technical community, developing the Fisheries Integrated Toolbox, and assisting in research to advance the NOAA stock assessment enterprise and improve regional communication

Communications Committee Chair

Oct 2022 - June 2026

Committee Member

June 2015 - Oct 2022

Association of the Marine Laboratories of the Caribbean

Manage internal and external communication for the AMLC, a nonprofit which connects marine laboratories, scientists, students, and stakeholders in the Wider Caribbean Region

North Pacific Groundfish Observer

June 2016 – Jan 2018

Saltwater Inc. / NOAA Fisheries

Collected data on catch, bycatch, and fishing effort on board commercial Alaskan fishing vessels to be used by the National Marine Fisheries Service to monitor fish populations, set catch limits, and manage the commercial Alaskan fisheries

PUBLICATIONS

Brodziak, J., M. Supernaw, B. Li, C. Stawitz, **H. A. Oleynik**, T. A'mar. (2025) Engineering a prototype for a next generation stock assessment modeling platform. *Fishery Bulletin*. 123(2). 88-104.

Oleynik, H.A., J.J Bizzarro, E.A. Hale, A.B. Carlisle. 2024. Environmental drivers of biogeography and community structure in a Mid-Atlantic estuary. *Oecologia*. <https://doi.org/10.1007/s00442-023-05500-Z>

Zahner, J.A, M.A. Heller-Shipley, **H.A. Oleynik**, S.G. Beyer, P-Y. Hernvann, M. Véron, A.N. Odell, J.Y. Sullivan, A.L. Hayes, V.G. Gertseva, K.L. Oken, O.S. Hamel, M.A. Haltuch. 2023. *Status of Shortspine Thornyhead (Sebastolobus alascanus) along the US West coast in 2023*. Pacific Fishery Management Council, Portland, Oregon. 139 p. www.pcouncil.org/shortspine-thornyhead

Blackhart, K., and **H.A. Oleynik**. 2023. *Setting targets and analyzing data gaps for U.S. fish assessments*. NOAA Tech. Memo. NMFS-F/SPO-242, p. 77 <https://repository.library.noaa.gov/view/noaa/51033>

Oleynik, H.A. 2022. *Living Resources: Fish Population Trends in Delaware Bay: Climate and Community*. Technical Report for the Delaware Estuary and Basin, J. Shinn, L. Haaf, L. Morgan, and D. Kreeger (eds). Partnership for the Delaware Estuary. <https://delawareestuary.org/data-and-reports/state-of-the-estuary-report-2/>

ICES (2022): Working Group on Technology Integration for Fishery-Dependent Data (WGTFD; outputs from 2021 meeting). ICES Scientific Reports. Report. <https://doi.org/10.17895/ices.pub.19367885.v1>

Mathews, J.A., E.A. Hale, **H.A. Oleynik**, et al. (2022) *Exploring Trends in Abundance of Young-of-the-Year and Age-1 Atlantic Croaker, Black Drum, Spot, and Weakfish in Relation to Salinity, Temperature, and Large-Scale Climatic Signals in a Mid-Atlantic Estuary*. *Transactions of the American Fisheries Society*, 151(2), 150–171. <https://doi.org/10.1002/TAFS.10332>

PRESENTATIONS

- "Impacts of Dam Operations on Mortality and Injury of Juvenile Chinook Salmon at Cougar Dam"* Willamette Science Review 2026
- "Population-level survival impacts of Harbour seal foraging on juvenile Pacific salmon, and the mechanistic links"* PICES 2025
- "Drivers of Productivity in Fraser River Salmon"* American Fisheries Society WA-BC Chapter Meeting 2025
- "Incorporating Human Dimensions into EBFM: Understanding Socioeconomic Consequences of Ecological Change"* American Fisheries Society Annual Conference 2024
- "Integrating Electronic Monitoring Data into U.S. Fish Stock Assessments"* World Fisheries Congress 2024
- "Fisheries Science and Management"* Guest Lecture, George Washington University Marine Ecology Course (BISC 3454) Dec 2024
- "Long-Term Shifts in the Fish and Macroinvertebrate Community in a Mid-Atlantic Estuary"* American Fisheries Society Annual Conference 2021
- "Changes in a Mid Atlantic Estuary: Long Term Shifts in the Fish and Macroinvertebrate Community in Delaware Bay"* NOAA Central Library 2021
- "Long-Term Shifts in the Fish and Macroinvertebrate Community in a North American Estuary"* World Fisheries Congress 2021
- "Changes in a Mid-Atlantic Estuary: Long-Term Trends in Environmental Conditions and Marine Community in Delaware Bay"* American Fisheries Society Mid-Atlantic Chapter Meeting 2020
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AWARDS AND FUNDING

- Partnership Engage Grant** Social Sciences and Humanities Research Council, Canada 2024
- Buck Suzuki Legacy Bursary Award** T. Buck Suzuki Foundation 2022
- Frances Severance Thesis Distinction Award** University of Delaware 2021
- Daiber Fellowship** University of Delaware 2019
- Women of Promise Award** University of Delaware 2019
- Okie Fellowship** University of Delaware 2018
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OUTREACH AND ENGAGEMENT

<i>REDI Committee</i> Student rep , UBC Institute for the Oceans and Fisheries	2022-present
<i>Society for Women in Marine Science</i> Co-Chair , University of Delaware Chapter	2019-2020
<i>Graduate Student Association</i> Co-Chair , UD School of Marine Science and Policy	2019-2020
<i>Hutton Junior Fisheries Biology Internship</i> Reviewer , American Fisheries Society	2024
<i>John A. Knauss Marine Policy Fellowship</i> Reviewer , NOAA Sea Grant	2022, 2024

EXPERTISE

Computing

Advanced in R, WinBUGS, JAGS, Git/GitHub; experience with C++, Stan, Stock Synthesis, Template Model Builder (TMB), VAST

Field & Lab

AAUS Scientific Diver, SDI Rescue Diver, small boat operation, random field sampling, otolith collection, shark tagging, marine animal husbandry
